

$$2 \quad (a) \quad \text{ps} = 10^{-12}(\text{s}) \text{ or } T = 4 \times 50 \times 10^{-12}(\text{s}) \quad \text{B1}$$

$$v = f\lambda \text{ or } v = \lambda / T \quad \text{C1}$$

$$\lambda = 3.0 \times 10^8 \times 4 \times 50 \times 10^{-12} \quad \text{C1}$$

$$= 0.06(0)\text{m} \quad \text{A1} \quad [4]$$

$$(b) \quad 1500 = 3.0 \times 10^8 \times 4 \times \text{time-base setting} \text{ or } T = 5 \times 10^{-6}\text{s} \quad \text{C1}$$

$$\text{time-base setting} = 1.3 \text{ (1.25)} \mu\text{s cm}^{-1} \quad \text{A1} \quad [2]$$